



Methods of Inquiry

The Questions We Ask Determine the Answers We Find

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Ordinary Human Inquiry

Individual approaches to explaining things.

OHI is similar to other methods of inquiry...

- OHI is causal and probabilistic (we learn that actions have consequences and some things are more likely to happen than others)
- OHI aims to predict outcomes by making observations and recognizing patterns

BUT OHI is less rigorous and systematic in understanding probability and causality than scientific inquiry.

OHI often involves prediction *without* understanding...

- We might have an idea of *what* will happen (prediction) without understanding *why* it does (that's pretty weak)
- Scientific inquiry provides opportunities for prediction AND understanding (which is much stronger).

Individual experiences.

Second-hand knowledge (agreement reality):

- Tradition
- Authority

Traditional Knowledge

What should you do if you get stung by a jellyfish?

Don't swim after doing what?

You shouldn't keep making that face or else what?

What color do bulls hate?

What do you do if you drop your food on the floor?

{Blank} stunts your growth.

How much of our brains do we use?

Individual experiences.

Second-hand knowledge (agreement reality):

- Tradition (e.g. old wives' tales)
- Authority (e.g. my mother)

OHI: Sources of Knowledge

Individual experiences.

Second-hand knowledge (agreement reality):

- Tradition (e.g. old wives' tales)
- Authority (e.g. my mother)

BUT...

Errors in Inquiry

We also make frequent errors in inquiry and logic.

Some errors in inquiry:

1. Inaccurate Observations
2. Overgeneralization
3. Selective Observation
4. Illogical reasoning
5. Ideology and politics

Come up with some examples of these errors from real life or some hypothetical scenario.

Errors in Inquiry

We also make frequent errors in inquiry and logic.

Some errors in inquiry:

1. Inaccurate Observations (casual vs. purposeful obs.)
2. Overgeneralization (limited vs. extensive obs.)
3. Selective Observation (ignoring evidence that contradicts)
4. Illogical reasoning (using faulty logic to draw conclusions)
5. Ideology and politics (being a hedgehog)

Foundations of scientific inquiry:

- More formal logic (understanding must make objective sense)
- Empirical observation (understanding must be consistent with what is consistently observable)

Goals of scientific inquiry:

- discover relationships between variables
- discover patterns that allow us to describe, predict, or prescribe.